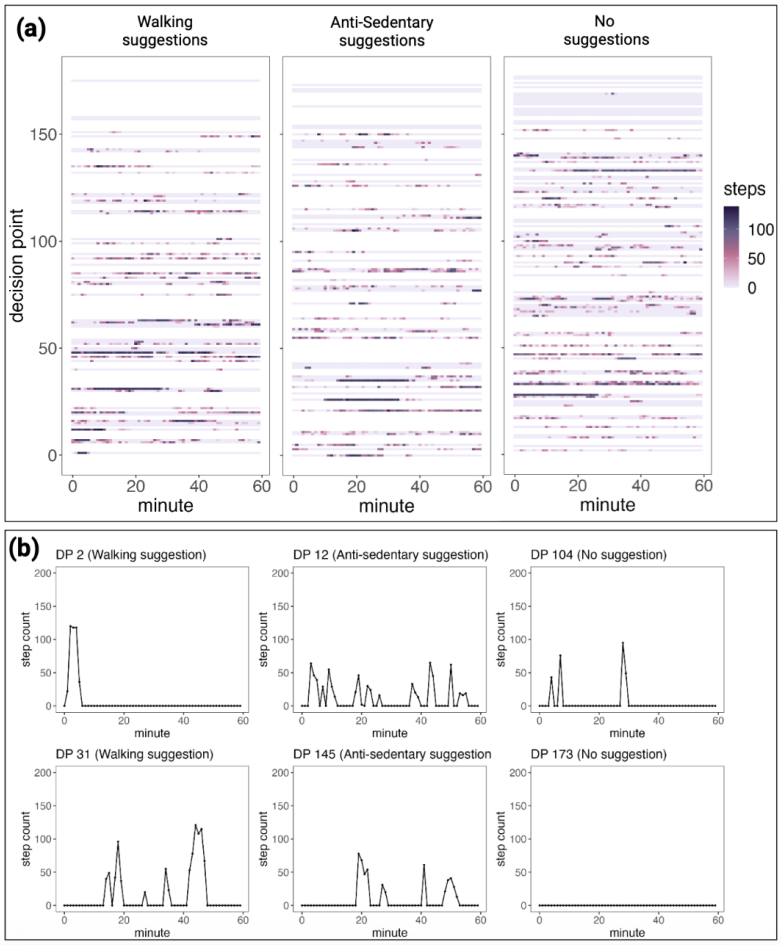

Figure 1: (a) The minute-level step count over the 60-minute window following each of the 210 decision points for one participant. Decision points with different types of suggestions are separated into three panels. Darker color indicates higher step count in that minute. (b) The minute-level step count from six selected decision points for that participant.

categorical treatments, with derivations in Section D of the Supplementary Material.

In intervention design, researchers might consider it unsafe or unethical to send push notifications at certain decision points, such as during driving. This concept is termed "(un)availablility" in the literature (Boruvka et al., 2018). Formally, X_{ij} includes an indicator I_{ij}, with $I_{ij} = 1$ denoting being available at decision point j and $I_{ij} = 0$ otherwise. If $I_{ij} = 0$, then $A_{ij} = 0$ deterministically.

The overbar is used to denote a sequence of variables up to a decision point; for example, $\bar{A}_{ij} = (A_{i1}, ..., A_{ij})$. Information accrued up to decision point j is represented by the history $H_{ij} = (X_{i1}, A_{i1}, Y_{i1}, ..., X_{i,j-1}, A_{i,j-1}, Y_{i,j-1}, X_{ij}) = (\bar{X}_{ij}, \bar{A}_{i,j-1}, \bar{Y}_{i,j-1})$. The randomization probability for A_{ij} is a constant in HeartSteps but can generally depend on H_{ij} and we denote it by $p_{ij}(H_{ij}) = P(A_{ij} = 1|H_{ij})$. We assume that the data from different individuals are independent and identically distributed samples from an unknown distribution. Throughout, we represent random variables or vectors with uppercase letters, and their realized values with lowercase letters.

In HeartSteps, each participant was in the study for $J = 5 \times 42 = 210$ decision points. X_{ij} records information such as the individual's location and the current day of the week. A_{ij} represents whether an activity suggestion is sent to individual i at decision point j, randomized with a constant probability 0.6 when $I_{ij} = 1$. $I_{ij} = 0$ if the individual is driving or already walking. We examine the effect of the activity suggestion over the subsequent 60 minutes, thus $T = 60$ and $Y_{ij} = (Y_{ij1}, ..., Y_{ij60})^T$.

We use $[n]$ to denote the set $\{1, 2, \ldots, n\}$. For reference, a list of symbols used is summarized in Section F of the Supplementary Material.

3.2 Potential Outcomes and Causal Effects

To define causal effects, we use the potential outcomes framework (Rubin, 1974; Robins, 1986). Specifically, let $X_{ij}(\bar{a}_{i,j-1})$ be the observation that would have been observed had that individual received the treatment sequence $\bar{a}_{i,j-1}$. Let $Y_{ij}(\bar{a}_{ij})$ be the collection of potential outcomes $\{Y_{ij1}(\bar{a}_{ij}), ..., Y_{ijT}(\bar{a}_{ij})\}^T$ under treatment sequence $\bar{a}_{ij}$.

mHealth interventions are typically designed to influence an individual proximally in time. For example, the activity suggestions in HeartSteps aimed to urge individuals to stand up and walk in the moment. We define the causal excursion effect (CEE) of the treatment A_{ij} on the proximal outcome Y_{ijt} to capture such impacts:

$$\text{CEE}\{j, t, S_{ij}(\bar{A}_{i,j-1})\} = \mathbb{E}\{Y_{ijt}(\bar{A}_{i,j-1}, 1) - Y_{ijt}(\bar{A}_{i,j-1}, 0) \mid S_{ij}(\bar{A}_{i,j-1}), I_{ij}(\bar{A}_{i,j-1}) = 1\}, \quad (1)$$

where $S_{ij}(\bar{A}_{i,j-1})$ is a vector of summary variables formed from $H_{ij}(\bar{A}_{i,j-1})$ based on researchers' analysis goal. Expression (1) represents the contrast between the expected outcomes at (j, t) under two treatment policies that are viewed as "excursions": following the MRT policy until decision point $j-1$, then either receiving treatment at decision point j or not receiving treatment at decision point j. The causal effect is marginal over the distribution (under the MRT policy) of $\bar{A}_{i,j-1}$ not included in $S_{ij}(\bar{A}_{i,j-1})$. The effect is conditional on the moderators of interest $S_{ij}(\bar{A}_{i,j-1})$, and different choices of $S_{ij}(\bar{A}_{i,j-1})$ can address different scientific questions. For example, setting $S_{ij}(\bar{A}_{i,j-1}) = \emptyset$ yields a fully marginal effect. Setting $S_{ij}(\bar{A}_{i,j-1}) = \text{loc}_{ij}$ yields causal effect moderated by the real-time location of the participant, loc_{ij}. The effect is conditional on availability $I_{ij}(\bar{A}_{i,j-1}) = 1$; i.e., we only focus on available times, because it is unethical or impractical to send interventions when someone is unavailable. Expression (1) generalizes the CEE in Boruvka et al. (2018) and Qian et al. (2021) to longitudinal functional outcome: for a fixed decision point and a moderator value,